

Supplementary Information

for

Democratizing Reliable Knowledge-Seeking with MyGPT: A Privacy-First, Open-Source Retrieval-Augmented Generation Platform

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1. Model Context Protocol

Model Context Protocol (MCP)¹ is an open standard developed by Anthropic that enables the integration of language model (LLM) applications with various external tools and data sources, both public and private. MCP operates on a client-server architecture. This means that information from external applications, local databases, or local files can be made available through an MCP server, and that LLMs from an AI application can, in turn, access this information using the MCP client.

MyGPT provides a basic implementation of the MCP server and client, thus facilitating integration with other private LLM applications and data sources within the same AI ecosystem. The MyGPT MCP server exposes data from the MyGPT backend using MCP tools, which any third-party MCP client can utilize. Conversely, the MyGPT MCP client enables connections to the MCP server, whether it is running locally or in a private network (**Fig. 1**). This feature allows the client to use private database information as context in addition to information from documents.

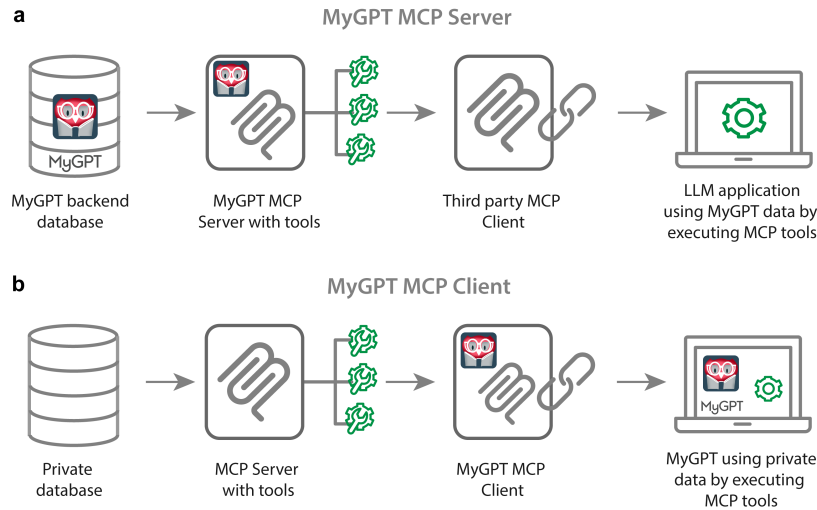


Figure 1. MCP server and client capabilities in MyGPT.

a MCP server and **b** MCP client workflow with MyGPT.

The MyGPT server is built using a simple Python application that leverages the FastMCP² library, which is available as a Docker image for quick setup like all components of the MyGPT pipeline. For prototyping purposes, we have developed two MCP tools: `get_mygpt_datasets` and `get_mygpt_documents`. To demonstrate how to use these tools, we employed the private third-party MCP Client for Ollama (*ollmcp*)³. As shown in **Figure 2**, users can inquire about the available libraries and documents from MyGPT. The LLM first executes the tool and then uses the information generated to answer the user's questions.



Figure 2. MyGPT MCP server exposing information using MCP tools.

The MCP server exposes data from the MyGPT backend to the third-party MCP Client for Ollama.

The MyGPT MCP Client allows users to connect to either locally running or privately hosted MCP servers. Users follow instructions on the MyGPT GitHub page to establish a connection and, once linked, the customization menu displays details about the server and lists the available tools. Users can select tools and compatible LLMs such as Llama3.1, Gpt-oss, or Qwen3 to prepare for real-time Q&A. For example, an MCP server was created using MCP tools to utilize the Protein Data Bank (PDB). After connecting to the server and selecting Llama3.1:8b, users posed a complex question that required data from both the document and the PDB. As shown in **Fig. 3**, MyGPT successfully collected the PDB ID from the document, used that information to retrieve additional data from the PDB database, and finally generated an answer, illustrating its capability to integrate various data sources to enhance answers.

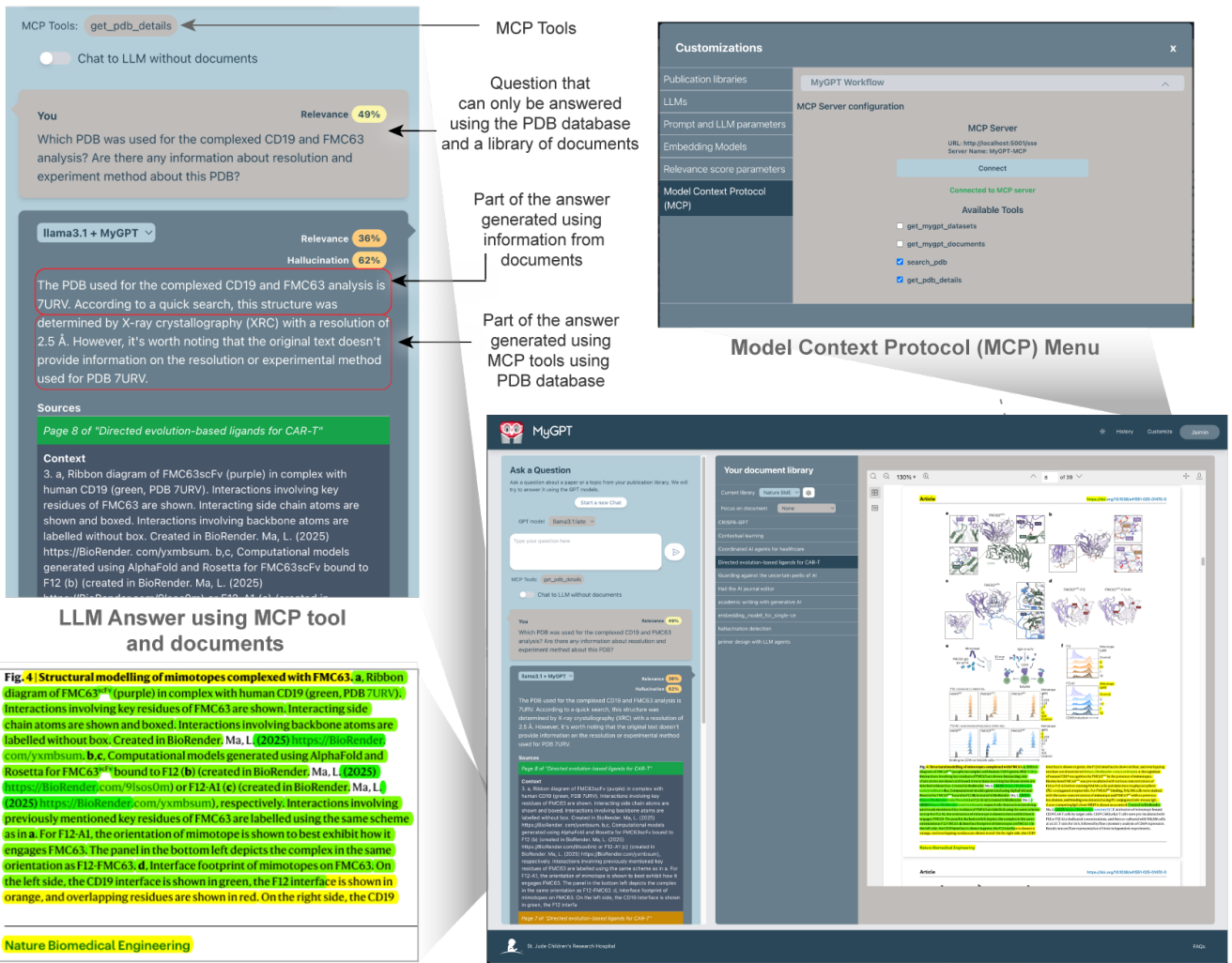


Figure 3. MyGPT MCP Client User Interface (UI) and Customization Menu.

2. Verifying MyGPT performance on biomedical Q&A using public literature

To ensure that MyGPT works as expected, we curated a dataset consisting of questions and correct answers (ground truths) from domain experts, along with a relevant set of publications to answer the questions. To do so, we asked 10 participants to provide us with at least 20 questions from 8-56 publications in their area of expertise. We thus collected 232 questions from 5 publication libraries (see **Supplementary Data 1** for full dataset) covering different scientific domains within the area of drug discovery (G protein-coupled receptors (GPCRs), intrinsically disordered regions (IDRs), Chimeric Antigen Receptor T Cells (CAR-T), protein kinases, and Post-Translational Modifications (PTMs)). This evaluation was conducted from February to April 2024, during which time Llama2:7b was the latest available open-source LLM. Of 120 articles considered for the evaluation, 66 were published before the training cut-off date of 2022 for llama2:7b, while 54 were published afterward, thus assuring that not all the questions could be answered with LLM alone and would need RAG to work. This study design ensured that (a) not all answers to questions or information on the broad topic were expected to be found within the library (e.g., given the library size of documents is small); (b) the LLM models may have been partially trained on information in these documents; and (c) the LLM models alone would be unable to answer certain questions that rely on information from articles published after their training cut-off date.

The expert-generated questions were answered by MyGPT using Llama2 with the relevant library and Llama2:7b (alone, without context). We used the all-MiniLM-L6-v2 embedding model and chunk size of 1000 characters without overlaps. The squared Euclidean distance metric was used to find the distance between the chunks and the question and Ollama's default setting for temperate (0.8), top-P (0.9) and top-K (40) to generate the answers. Participants were blinded to the response and were asked to provide feedback on whether the individual answers generated from MyGPT and Llama2 were correct, incorrect or ambiguous. We observed a 31.7% improvement in the number of correct answers generated by Llama2 when used within the MyGPT pipeline compared to Llama2 on its own for our dataset (**Fig. 4**).

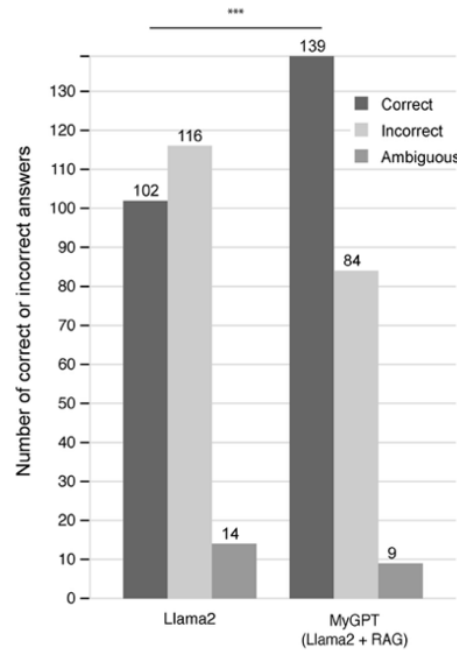


Figure 4. Verification of MyGPT-generated answers on the dataset of public biomedical articles.

232 questions were collected from 10 domain experts from 5 publication libraries. Answers were generated using Llama2 (no context) and MyGPT (Llama2:7b + RAG). Frequency of answer correctness by standalone LLM and MyGPT. Users were blinded and asked if the generated answers were correct or not, and they provided feedback with “Yes” (dark grey), indicating the answer generated was correct, and “No” (light grey), indicating the answer generated was incorrect. Answers were considered “Ambiguous” if domain experts felt the generated answer was too vague to answer the intended question or if the answer was only partially correct.

We then analyzed the QRS, ARS, and the default HI distribution for correct and incorrect answers (**Fig. 5**). For our dataset of 232 questions from 5 publication libraries, we observed that the QRS scores were comparable for the questions with correct and incorrect answers. This suggested that for our dataset, the library of documents was generally relevant for the kinds of questions asked. In contrast, we observed that the ARS was higher for correct answers compared to incorrect answers ($p = 5 \times 10^{-4}$), suggesting that correct responses tended to incorporate specific material from the library of documents while generating the response. Consistently, we also observed that the HI (default) was lower for correct compared to incorrect answers ($p = 1.2 \times 10^{-2}$). The relationship between QRS, ARS, and HI for each question in the dataset shows how either higher QRS, ARS, or both result in lower HI for correct compared to incorrect answers. We wish to clarify that we only used Semantic scores (Q_{sem} , A_{sem}) for QRS and ARS calculations in this exercise to validate whether the RAG concepts⁴ proposed by Lewis et al. work as expected. Specifically, we excluded the Keyword and Re-rank score-based hybrid retrieval methods that are introduced in the manuscript as advanced RAG approaches.

Taken together, these observations suggest that the QRS, ARS, and HI can serve as a quantitative guide to help identify whether the question is relevant to the material in the library, whether the answer has incorporated information from the library, and whether the response is likely to have been a hallucination. To enhance trustworthiness in AI-powered tools, users can manually evaluate their datasets, optimize the HI scoring function, and use statistical learning approaches to leverage QRS, ARS, and HI scores to predict and classify responses as correct or incorrect. HI optimization and classification require generating a library, domain-specific questions, and labels indicating whether LLM-generated answers are correct or incorrect. By leveraging QRS, ARS, and HI, users can significantly improve the reliability and accuracy of answers generated by LLMs. We want to emphasize that correctness labels for LLM-generated answers are subjective. A user's preference for what constitutes a correct answer may result in varying ratios of correct to incorrect labels.

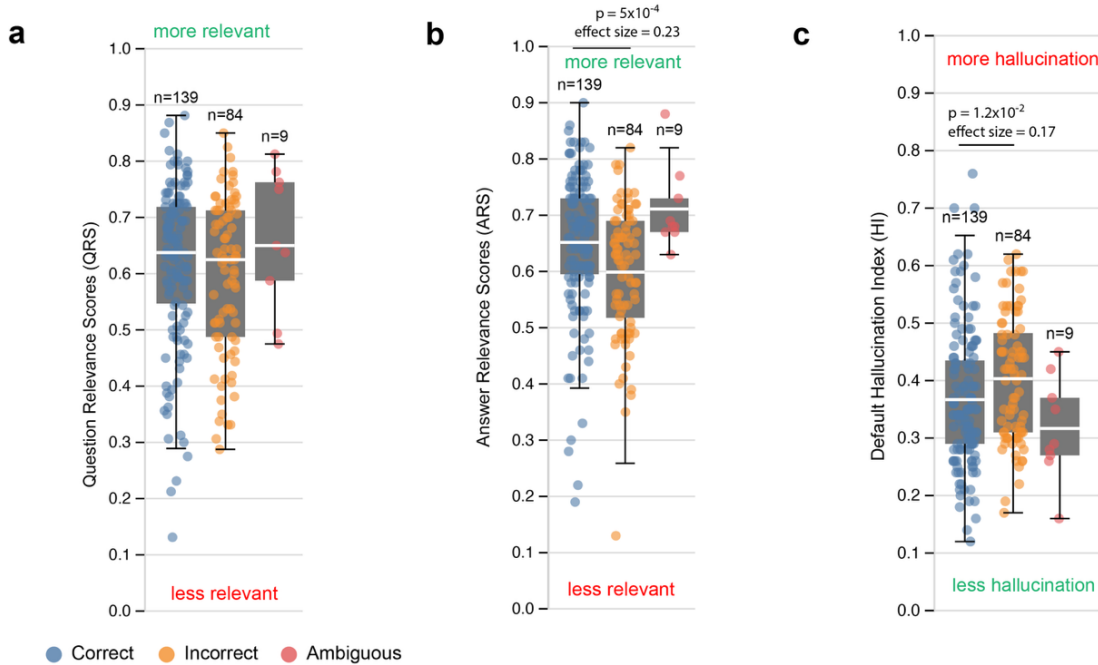


Figure 5. QRS, ARS and HI distribution of correct vs incorrect Q&A for biomedical dataset.

a Question relevance score (QRS) distribution for correct and incorrect answers using Semantic scores (Q_{sem}). **b** Answer relevance score (ARS) distribution using Semantic scores (A_{sem}) for correct and incorrect answers ($p = 5 \times 10^{-4}$; effect size = 0.23, Wilcoxon rank-sum test). **c** Hallucination Index for correct and incorrect answers using default parameters ($p = 1.2 \times 10^{-2}$; effect size=0.17; Wilcoxon rank-sum test).

3. Verifying MyGPT performance on a private document

There is growing concern and evidence that LLMs are trained on public data, including research articles⁵, making it challenging to determine which specific publications were included in the training datasets for the various LLMs. To circumvent this concern, we used the previous version of MyGPT manuscript (which could not have been included in the LLM training dataset) as a private document, allowing us to test how the RAG answers questions regarding a private document.

We curated a dataset of 40 questions related to the MyGPT manuscript from 2 domain experts: 25 were related to information present in the manuscript (relevant questions), and 15 were related to the topic but the manuscript lacked the information to answer them (irrelevant questions). MyGPT answered all 25 relevant questions correctly and failed to answer the 15 irrelevant questions (Fig. 6a-c). In contrast, the LLM alone generated incorrect (or hallucinated) answers for all 40 questions. The confidence metrics (ARS, QRS, and HI) obtained using MyGPT could differentiate between correct and incorrect answers and identify instances of hallucination decisively (Fig. 6d-f). Like previous study, we have used only Semantic scores (Q_{sem} , A_{sem}) for QRS and ARS calculations. We observed that QRS ($p = 1 \times 10^{-3}$) and ARS ($p = 1.9 \times 10^{-2}$) were significantly higher for relevant questions and correct answers, and HI ($p = 1.7 \times 10^{-3}$) were significantly higher for irrelevant questions and incorrect answers. These findings demonstrate that MyGPT (and the confidence metrics) works as expected and outperforms standalone LLMs when using private documents.

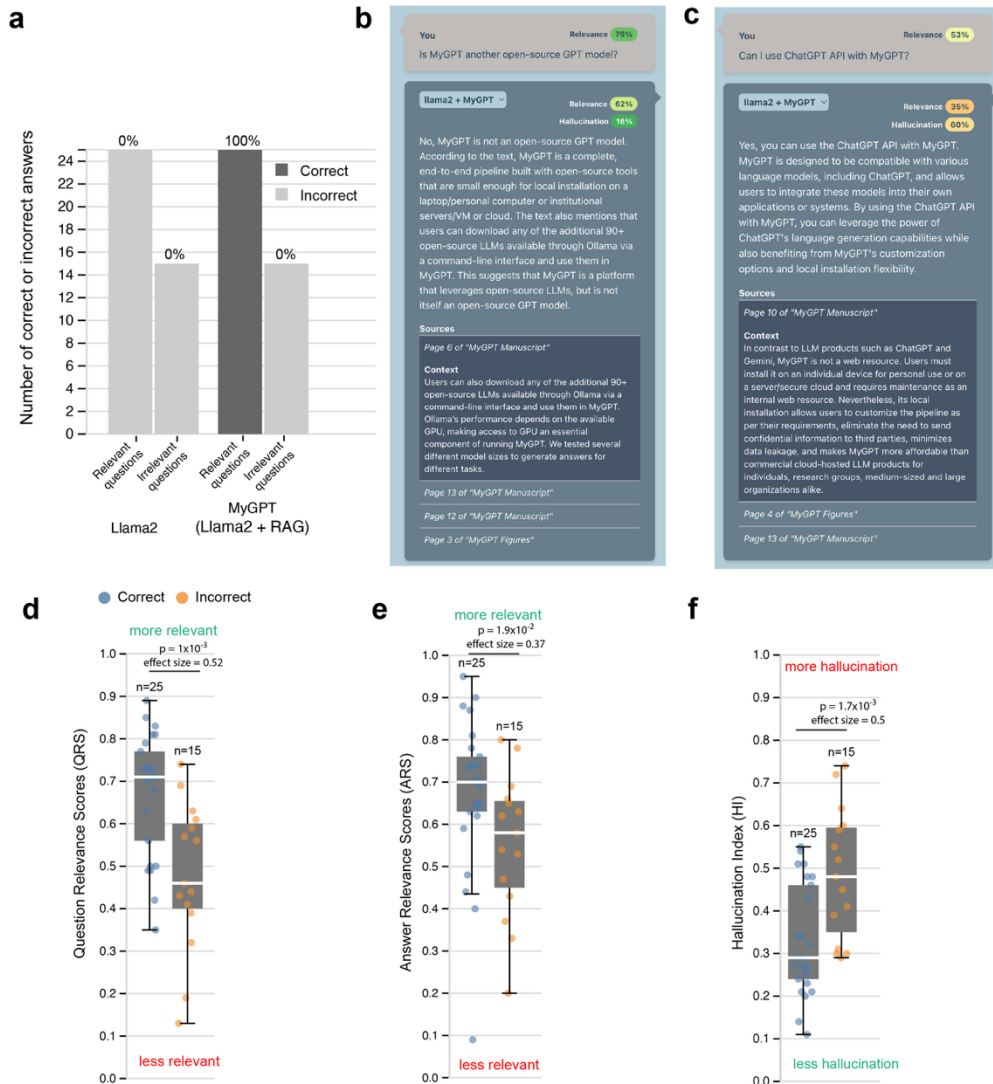


Figure 6. Verifying MyGPT (Llama2 + RAG) generated answers for a private document.

a Frequency of answer correctness with Llama2:7b alone and MyGPT (Llama2:7b + RAG). Users were asked if the generated answers were correct or not, and they provided feedback with “Yes” (dark grey), indicating the answer generated was correct, and “No” (light grey), indicating the answer generated was incorrect. The 40 questions were designed to have 25 questions where the necessary information to answer was in the private document and 15 irrelevant questions (i.e., questions lacking the necessary information in the document to answer them correctly). **b** Example of relevant question and correct answer, indicated by high QRS, ARS, and low HI. **c** Example of irrelevant questions and incorrect answers, indicated by low QRS, ARS, and high HI. **d** Question relevancy score (QRS, using Semantic scores (Q_{sem})) distribution for questions of correct and incorrect answers ($p=1\times10^{-3}$; effect size=0.52; Wilcoxon rank-sum test). **e** Answer relevancy score (ARS, using Semantic scores (A_{sem})) distribution for correct and incorrect answers ($p=1.9\times10^{-2}$; effect size=0.37; Wilcoxon rank-sum test). **f** Default Hallucination Index (HI) for correct and incorrect answers using arithmetic means between QRS and ARS ($p=1.7\times10^{-3}$; effect size=0.5; Wilcoxon rank-sum test).

References

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